AT-01

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Sub:- Data structure

Code : CSA0306

1 An operating system manages dynamic memory dynamic memory allocation. Analyze how linked lists can represent free memory blocks. compare this with array based memory management in term of flexibility, fragmentation and performance, and justify the better approach.

Ans memory management is an important function of an operating system. It allocate memory to Process and release it after execution. A linked list is commanly used to represent free memory block Each node is the linked list stores the starting address size of the free block and a pointer to the next free block. when memory is allocated, the operating system finds a suitable blocks and update the linked list when memory is released the block is added back and adjacent free blocks are merged.

In an array-based memory management system, memory blocks are stored in fixed positions. Array blocks are stored are provides fast access using indexes, but insertion and deletion are difficult because element may need to be shifted.

Array also have a fixed size, making them less flexible.

Comparision

Linke List	Array
• Dynamic size.	• Fixed size.
• Easy insertion and deletion	• Difficult insertion and deletion.
• Better memory utilization	• may waste memory.
• slightly slower access.	• Faster access.

Advantages of linked list

- Dynamic memory allocation.
- Easy insertion and deletion of memory blocks.
- Better utilization of available memory.
- suitable for continuously changing memory requirement.

Advantages of array

- simple implementation
- Fast random access using index.
- suitable when memory size is fixed
- less pointer overhead

Justification

Linked list are suitable for operating system because they support dynamic memory allocation reduce memory wastage, and handle memory allocation and deallocation efficiently.

4 A system search records in a sorted dataset Compare linear search and binary search using time complexity and number of comparisons Evaluate which is more efficient and justify your answer

* Searching is the process of finding a required element in a dataset. Two common searching methods are linear search and Binary search

Linear search.

It checks each element one by one until the required element is found. It works for both sorted and unsorted data but becomes slow for larger dataset. It is time complexity is O(n) is the average and worst cases.

Algorithm

- starts from the first element
- Compare each element with the targent
- If found, return the position.
- otherwise, continue until the last element.
- If not found, report failure.

Binary Search

It works only on sorted data. It compares the middle element and repeatedly divides the data set in to two halves until the

element is found. Its time complexity is O(log n) making it much faster than linear search.

Algorithm

- Find the middle element.
- Compare the middle element with the target.
- If equal, the search is successful.
- If the target is small, search the left half.
- If the target is larger, search from right half
- Repeat until the element is found.

Comparision

Feature	Linear search	Binary search
• Data Required	• sorted or unsorted	sorted only.
• Worst case	• O(n)	O(log n)
• Comparisons.	• Upto n.	About $\log_2 n$
• Speed.	• Slower.	Faster.

Justification

Binary search is more efficient because it requires fewer comparision and less execution time. Linear search may check all records. while Binary search needs only about 10 comparisions. Therefore binary search is best for searching sorted dataset